GlucoSentinel: system architecture of the type 2 diabetes digital twin

Two data streams in, one calibrated virtual patient in the middle, a 2-hour spike warning and a doctor console out

1 Data sources

Static stream: EHR

Synthetic EHR
120 Synthea-style patients

HL7 FHIR R4 bundle
- Condition, ICD-10 **E11.9**
- Labs, LOINC (HbA1c **4548-4**)
- Medications, RxNorm
- TCF7L2 genotype + PRS

Dynamic: wearables

- **CGM glucose** every **5 min**
- **Smartwatch** heart rate, **HRV RMSSD**
- **Sleep stages** light, deep, REM, awake
- **Steps** per 5 min, walks
- **Meal log app** **32** Indian dishes

2 Ingest

FHIR parser
bundle to EHR features

5-min grid
align all streams

Sensor QC
gaps, dropouts

Sleep debt score
per night

Meal events
carbs split by GI

3 Digital twin engine

A. EHR prior HbA1c, FPG, BMI, duration, genotype, meds set starting physiology

B. Calibrate 7-day onboarding: fit insulin sensitivity, beta-cell, basal, dawn

C. Assimilate nudge the twin to each CGM reading every 5 min

D. Simulate 2-hour forecast and what-if: meal swap, walk, sleep, meds

Physiology core (per patient)

Glucose dG/dt = -(SG + X)G + SG Gb(t) + Ra(t)/V - ex steps G
Insulin action dX/dt = -p2 X + p2 SI s(t) (I - Ib)
Insulin dI/dt = -n (I - Ib) + gamma max(G - Gb, 0)
Gut fast and slow carb pools by meal GI
Modifiers dawn effect, stress, **sleep debt** lowers SI s(t)
Learned per patient SI, gamma, Gb, dawn.
Recovery of true insulin sensitivity: **r = 0.92**

4 Fusion and AI

81 fused features every 5 min

E 23	EHR
W 40	Wearables + CGM
R 5	Personal routine
T 13	Twin state

LightGBM classifier: P(spike above 180 in next 2 h)

4 LightGBM forecasters: +30, +60, +90, +120 min

SHAP: top drivers per alert

Alert policy: threshold set on validation, at most 1 per hour

5 Doctor and patient

Patient nudge

Spike likely after lunch

Try bajra roti instead of rice

Walk 20 min after eating

Doctor console
- Risk card and alert
- CGM plus 2-h forecast
- Why panel (SHAP)
- What-if simulator
- Weekly AGP, TIR
- FHIR record view

95 min	median warning lead
85%	spikes warned 30+ min ahead
0.82	false alerts per day

Governance, evaluation and stack

Data: synthetic only, no real patient data (DPDP Act, HIPAA sandbox)

Split: 84 / 12 / 24 patients, week 1 calibrates, week 2 evaluates

Checks: ablation, episode lead time, Clarke grid, subgroup AUROC

Stack: Python 3, NumPy, pandas, LightGBM, Streamlit, Plotly

Standards: HL7 FHIR R4, LOINC, ICD-10, RxNorm, ABDM ready

Licence: MIT open source; research prototype, not a medical device